

AI Article Reviews Dominate the Best Human Peer Reviews: Evidence and Policy Choices*

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Abstract

We show that AI-generated article reviews are now, by all relevant measures, better than the best human peer reviews. We then discuss how this result may inform journal policy choices, with potential paths to increase publication speed and quality; eliminate adversarial interactions; improve the experience of editors, reviewers, authors, and readers; and advance science.

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1 Introduction

Human peer review is the default quality-control standard for scholarly publishing, and yet problems with this system are deep, common, and well known (Aczel, Barwich, et al., 2025), including self-serving reviewer conduct, weak incentives and lack of expertise for some reviewers, and insufficient scrutiny; low inter-reviewer agreement (Bornmann, Mutz, and Daniel, 2010; Mutz, Bornmann, and Daniel, 2012); bias due to author characteristics, topics, and methods, with opacity and little accountability (Tennant and Ross-Hellauer, 2020); failure to detect major errors and little overall evidence of improved quality (Gasparyan and Kitas, 2012); frustrating delays (Farber, 2025); large costs in reviewer time (Aczel, Szaszi, and Holcombe, 2021); reviewer recruitment difficulties (Wright, 2024); unhelpful feedback (Chong and Lin, 2024); and resistance to heterodox findings (Armstrong, 1997).

These problems with the human peer review process have motivated scholars to evaluate large language models (now often called AI) as a replacement. But AI leads to other worries, including hallucinated claims, fabricated sources, and confidently stated errors (Ye et al., 2024); failure to recognize or reward novelty (Shin et al., 2025); biases learned from human training data unfairly favoring authors based on institution, seniority, or fame (Macharla Vasu et al., 2026; Hosseini and Horbach, 2023); excessive leniency (Russo Latona et al., 2024); vulnerability to manipulation (Lin, 2025; Ye et al., 2024); breaches of confidentiality (Naddaf, 2025); and unclear accountability for errors (Hosseini and Horbach, 2023; Porsdam Mann et al., 2025). The farthest scholars have gone so far is to conclude that AI should supplement rather than replace human peer reviews (Liang et al., 2024; Farber, 2025; Daskaliuk et al., 2025).

In this paper, we show through a novel data collection and head-to-head experiments that recent improvements in AI mean it now dominates the best human peer reviews — according to blind judgments by other AI models in every area and by (human) former editors of major scholarly journals. Given prior research on earlier AI models, we had expected AI to win some in some areas, some of the time, but we found that it wins in them all, by every criterion, by wide margins, and without the problems listed above. We

offer our research design in Section 2, statistical results in Section 3, and a discussion of implications for journal policy implications in Section 4.

2 Research Design

We now discuss the papers reviewed, the AI and human reviewers, and how we used AI and human judges to separately choose which review of each paper is better (see Supplementary Appendix A for more details).

Paper Submissions and Human Reviews For our study, we used papers submitted to *rapidPeer*, a service developed by Harvard University’s Institute for Quantitative Social Science (and then shared with several other institutions) that gives university faculty high quality double-blind reviews of their paper drafts prior to formal journal submission. The service asks human reviewers to read and comment on an entire paper and provide constructive suggestions throughout (as distinct from behaviors we sometimes see from journal reviewers, such as searching for a fatal flaw on p.3 so they can stop reading, or making it more about their own work than appropriate). Reviewers are told not to focus on whether the paper is suitable for a particular journal or meets a specific standard (although publication suggestions are welcomed) and instead asked to be comprehensive, to write at least 1.5-2 pages, and to try to be helpful to the author when revising. The instructions suggest a wide range of ways for reviewers to help (see Supplementary Appendix B for the exact language).

Reviewers are carefully chosen by experts in the field from institutions other than their own and incited to provide highly constructive reviews: They are paid (well) for their time and told they are likely to have another opportunity to be paid (or to submit a paper to the process themselves) if they do a good job. All manuscripts are deidentified prior to the review process. Whether judged by us, *rapidPeer* staff, or the paper authors (who filled out evaluations upon receipt), these reviews are indeed of exceptionally high quality. We also consulted former editors from major journals, all of whom concluded that the reviews are among the best they have seen in their times as editors.

The corpus we analyze comprises 124 scholarly paper submissions and 134 human reviews (10 submissions received two independent reviews) collected between 2019 and 2026 and drawn from a wide range of research, including statistical analyses of empirical data, formal mathematical modeling, and purely qualitative studies. Near as we can tell, all reviews were indeed written by humans: most predated ChatGPT; neither the quality nor style changed over time; and reviewers were compensated and responsible for their on contribution.

AI Reviews For each submission, we generated a single AI review with Claude Opus 4.8 (Anthropic, 2026) at its maximum reasoning setting, under two conditions. In the *matched* condition the model received the anonymized manuscript (including figures and tables) together with the verbatim *rapidPeer* reviewer guidelines (Supplementary Appendix B) as its instructions and no external tools, placing it in the same information environment as the human reviewer. In the *tuned* condition, the AI model received the same manuscript at the same maximum reasoning setting but with a purpose-built agent prompt tuned to the AI along with web retrieval abilities (Supplementary Appendix C). The tuned prompt was written before any judging, was never tuned against judges’ scores, and does not contain the judging rubric. Every tuned review invoked retrieval (mean 12.4 searches per review). Each review was produced in an isolated call — one review per submission per condition.

AI and Human Judges A model from a different family, GPT-5.5 (OpenAI, 2026), served as the AI judge, so that generator and judge share no lineage. Before comparing the two reviews, we had the AI decide which review topics were relevant to each paper (e.g., so a paper with no figures was not judged on “figures and tables”). We locked in that list of applicable topics once per paper, based only on the paper. Then we had the AI judge evaluate both reviews on only that set of topics. The AI judge saw the manuscript and evaluated the reviews under three instruments — an overall ranking of the two reviews, a head-to-head ranking of the two reviews in each of seven review areas, and an independent individual rating score from 0 to 10 for each of the areas (Supplementary Appendix D). Every comparison was run in both presentation orders, and a win was credited

only when the same review was preferred under both orders; the few order-inconsistent pairs, and pairs in which neither review addressed an area, were treated as undecided. Per-area win rates are AI wins divided by decided pairs, and rating values are means, each computed only over the submissions for which the area is relevant, so denominators vary across areas. The seven review judgement areas — theory and idea, methods, weaknesses, framing, communication, figures and tables, and sources — used the explicit *rapidPeer* reviewer guidelines rather than imposed externally (Supplementary Appendix E).

Finally, we enlisted three former editors of leading scholarly journals to independently judge 10 randomly selected and blinded review pairs, one AI and one human but ordered randomly, and under the same overall ranking instructions given to the AI reviewers. The editors were unaware they were comparing AI to human reviews (until we debriefed them afterwards). We did not attempt to ask our human editors to evaluate each review area separately (although debriefings with each editor revealed no differences in any area from the overall results). We collected this information via a web interface we designed for ease-of-use, with reviews for each pair presented side by side with independent scroll bars.

3 Results

Overall We now summarize in Figure 1 an analysis of (the 134 pairs of AI and human reviews, each of the same manuscript. The figure covers pairwise choices (Panel A), ratings of each item independently (Panel B), overall judgments (top), and AI ratings in seven review areas (bottom).

The results are unambiguous: No matter how we look at the data, AI wins. It wins in 100% of the pairs overall when judged by another AI model and 87% overall when judged by human journal editors (Panel A). It wins in 92% of AI-Human review pairs for three of the review areas (theory and idea, framing, and communication), 99% for one (figures and tables), and 100% in three (sources, weaknesses, and methods).

Although human coding has long been known to be notoriously error prone, we found that humans agreed as much or more with AI’s judgement than they did with other hu-

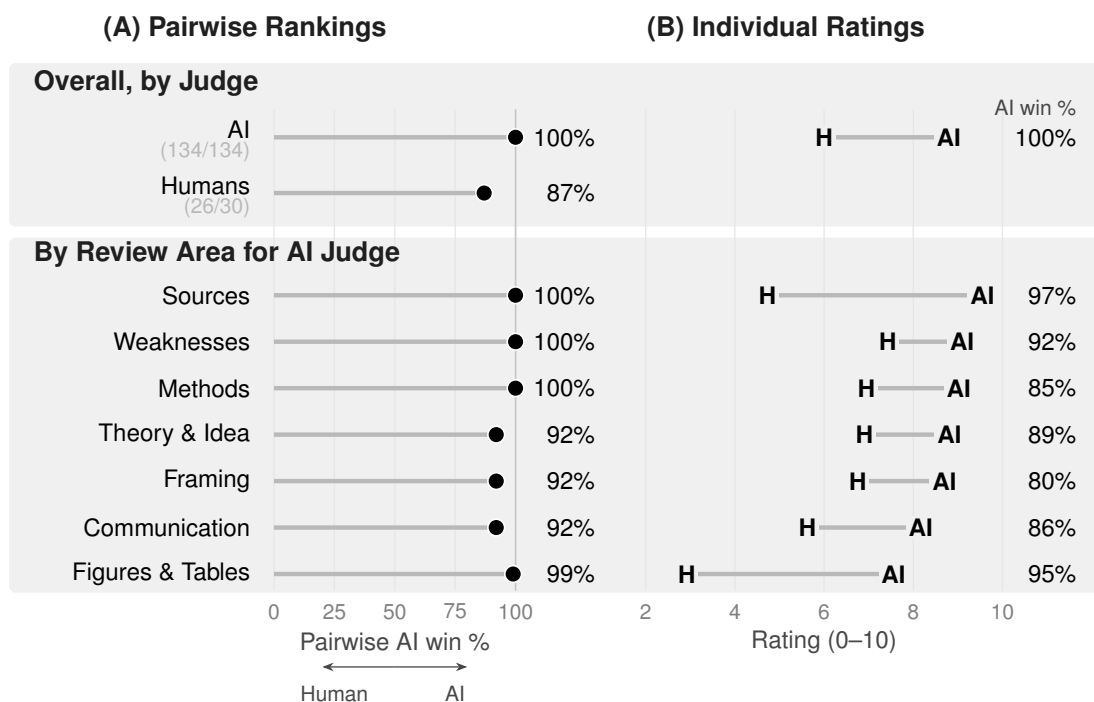


Figure 1: **AI reviews dominate human reviews.** The figure summarizes evaluations of 134 pairs of human and AI reviews of the same manuscript. The top reports overall evaluations by AI and by three (human) editors of major journals; the bottom summarizes results for seven separate review characteristics. Panel A gives pairwise rankings, the percentage of pairs where the AI review is ranked higher than the human review. Panel B gives mean ratings for both the human (**H**) and AI (**AI**) reviews.

mans: human editors agreed with other human editors 80% of the time but on average 87% of the time with the AI’s judgment.

Robustness Checks We also ran nine categories of robustness checks (Supplementary Appendix F): (1) To rule out the possibility that the AI wins merely by writing at greater length we restricted the overall comparison to pairs whose AI-to-human word-count ratio fell in [0.7,1.4]. On this length-matched subset of 44 pairs, AI was preferred in all 44 (100%), matching the full-corpus result. (2) AI wins even after biasing the sample by post hoc selection of the top human reviews, measured in various ways. (3) Human reviews are a high baseline according to the authors own evaluations: 88% judged the review they received (at the time they received it) as high or very high quality, and 100% indicated a willingness to recommend the rapidPeer service to other faculty. (4) AI vs human writ-

ing style (formatting or writing register) had no effect, which we showed by rewriting human reviews in AI style and finding no improvement in overall judgements. (5) The extent to which AI beats humans does not change over time, even after ChatGPT was released (and so AI judges that could have found published versions of submitted papers did not affect our conclusions). (6) Because reviews can be preferred even if incorrect, we checked accuracy in three ways and found that AI reviews did not hallucinate, were only rarely incorrect, and were correct more often than human reviews. (7) Despite the inherent stochasticity in AI reviews, a replication produced almost the same result. (8) The results above include the “tuned” AI reviews; the “matched” AI reviews perform almost the same, with the exception of winning by only 62% in the review area of “sources”. Once AI had access to a browser like the human reviewers did, it wins by 100%, as indicated in Figure 1. And finally, (9), we found no differences in the degree to which AI dominates human reviews by manuscript novelty, research style (statistical, mathematical, qualitative), faculty rank, age, and journal quality where the paper was eventually published.

4 Journal Policy Choices

High quality scholarly reviews are intended to help (1) authors improve their papers and (2) journal editors make publication decisions. Our results speak to the first of these two purposes: For constructive advice to improve scholarly work, our evidence supports the conclusion that AI reviews dominate the best human scholarly reviews, an advantage that is only likely to increase with advances in AI. Moreover, because editors rarely receive reviews as high quality as those we used for evaluation, the impact of using AI in the scholarly review process may be larger than even our results indicate. Our results do not speak to the second purpose, which is how editors make publication decisions, but we can use what we learned about (1) to inform journal decisions regarding both (1) and (2).

The policies of scholarly journals, like all policies, are at best informed, but never determined, by scientific findings, regardless of how strong or striking. Policies are decided by political processes, in this case by editors, editorial boards, and scholarly or other

organizations that control the journals. Thus, we offer here five ideas for journal policy changes that our results may inform, to the potential benefit of editors, authors, reviewers, readers, and science as a whole.

First, one simple policy change may be for journals to recommend or require that authors conduct repeated AI reviews themselves before submission, iterating with AI to improve their papers (via what are essentially self-supervised revise-and-resubmits), and continuing until authors are convinced they have resolved all resolvable issues and improved the paper as much as possible within time or resource constraints. (Authors should also be made aware that naive uses of AI makes individual scientists more creative than they would be without it but less than they could be, especially when considered for science as a whole (Hao et al., 2026; Wenger and Kenett, 2026), along with algorithms effective in fixing the problem (Luo et al., 2026).)

Second, each journal should consider adopting and publishing a specific AI prompt for authors to use for evaluation of their papers (for a starting point, we suggest combining the *rapidPeer*'s prompt in Supplementary Appendix B with the current "instructions to contributors" each journal already makes public). This policy will then use AI to better align submissions with each journal's specific editorial standards, reduce uncertainty in the review process, improve quality, and potentially even reduce rejections, especially desk rejects.

Third, journals should consider requiring the AI review (generated by the journal's official AI prompt) to be submitted with the paper, along with a letter from the author explaining any remaining open questions or disagreements with AI. The author's accompanying letter is crucial so that AI model creators do not get to determine the content of scholarly papers and because, as the paper improves during the iterations, AI sometimes becomes absurdly pedantic even though still technically correct (such as "yes $2 + 2 = 4$, but you never defined the $+$ sign").

Fourth, another step would be for journal editors to end human reviews and use AI reviews instead, as our evidence shows that AI reviews are better, and AI reviews alone have been shown to improve quality (Thakkar et al., 2026). However, making this one

change in isolation would ignore the equilibrium: if authors are aware of this change, they will likely use AI to review their papers themselves. Indeed, this decision may even already be a fait accompli since, despite the efforts of some editors to stop AI use in (supposedly human) reviews, it is becoming increasingly widespread (Naddaf, 2026).

Finally, a more consequential policy change that may also better respect the equilibrium outcome is for editors to skip the *routine* use of human reviews. Then the editor could review a dossier submitted by the author (consisting of the paper, the AI review generated from the journal's official AI prompt, and the author's response to the AI), a new AI review generated by prompts the editor chooses and does not shared publicly (such as including manipulation checks; Murray 2025), and, if necessary for particular cases, views of a chosen editorial board member or even, in difficult cases, an outside traditional human reviewer. The editor would then would makes a final publication decision.

This last idea, or some modification thereof, has the potential to greatly speed the process, enabling publication decisions within days rather than months. It would remove a lot of the bias and painfully adversarial parts of the process for the author (such as having to make unnecessary or sycophantic changes in papers and revise-and-resubmit responses to avoid offending a human reviewer). Papers submitted using iterative AI review will likely be more free of errors and weaknesses in logic, math, references, argument, scholarly literature, data analyses, graphic presentations, and even journal stylistic choices (AI obviously makes mistakes, although less often than ordinary human reviewers). Problems of special concern to each journal would be more easily avoided by including them in the journal's official AI prompt (as AI is considerably better at following long lists of detailed instructions to authors than humans tend to be).

The job of the editor, under this policy change, would be vastly improved and of interest to the high quality people the scientific community attracts for this job. Policy changes such as these may also help with the framing, impact, overall importance of the paper, and suitability for the journal, but these crucial decisions are, at the end of the day, the most important ones and so journals will presumably wish these to be left to the journal editor. Even if AI is perfect, authors are conscientious in iterating with AI to

improve their paper, and paper submissions are all correct or close to it, whether a correct paper meets the standards of any one scholarly journal can only be decided by the journal editor with the proper authority and responsibility.

We encourage others to follow up our research in other substantive areas, and with each new generation of AI models, to see how it might be used to continue to improve the scientific process and increase the speed at which science advances.

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